Isotopic evidence from the eastern Canadian shield for geochemical discontinuity in the Proterozoic mantle

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Most workers agree that Proterozoic anorthosite massifs represent the crystallization products of mantle-derived magmas[1,2], although the composition of the parental melts is a major unsolved petrological problem[3]. As mantle-derived rocks, the massifs can be used as geochemical probes of their late Precambrian upper mantle sources. We report here Nd and Sr isotopic compositions of anorthosites and related rocks from the Grenville and Nain Provinces of the eastern Canadian shield. Here 75% of the Earth's known anorthosite is found in a 1,600-km belt from the Adirondack Mountains of northern New York State to the eastern coast of Labrador[4] (Fig. 1). The results indicate that the massifs were derived from at least two distinct mantle source regions which were established before 1,650 Myr ago, and were episodically involved in magmatism over ~500 Myr. One reservoir, below the Grenville Province, and probably below much of the eastern Superior Province, was isotopically similar to the depleted, modern-day mid-ocean ridge basalt (MORB) source. The other reservoir was chondritic to moderately enriched, and is most easily identified in the Nain Province, but may have occurred scattered throughout the Superior Province.

The anorthosites were emplaced into presumed Archaean quartzofeldspathic gneisses, although in large areas of the Grenville Province, both massifs and basement rocks were reworked during late Proterozoic (about 1,000 Myr) Grenvillian metamorphism[5]. The Adirondack anorthosite may have crystallized as much as 300 Myr before high-pressure granulite facies metamorphism[5]. The other anorthosite massifs that we analysed are unmetamorphosed and include the large (7,000–8,000 km^2) composite massifs at Harp Lake[6] and Mealy Mountains, Labrador[7], and a smaller complex (300 km^2) at St Urbain, Quebec[8] (Fig. 1). Isotopic data have been published for three mafic intrusions in Labrador: the Kiglapait layered intrusion[9–11] and the Flowers River gabbro[12], both associated with the Nain anorthosite complex, and the Shabogamo gabbro, which intrudes supracrustal rocks of the Labrador Trough on both sides of, but within 25 km of the Grenville Front[13,14] (Fig. 1).

Sr and Nd isotopic data for these anorthositic and mafic intrusions are given in Table 1. The massifs can be divided into two types based on location and Nd isotopic signature (Fig. 2). This division is possible despite uncertainties of up to 200 Myr in age and five epsilon units in initial Nd isotopic ratio (ε_{Nd} as defined in ref. 15), caused by the effects of metamorphism and crustal contamination[5]. South-east of a line roughly parallel to the Grenville Front (Fig. 1) anorthosites have positive ε_{Nd} values, implying sources with long-term depletion in light rare earth elements (LREE). To the north-west, anorthosites and mafic intrusions in Labrador have negative ε_{Nd} values that suggest derivation from LREE-enriched sources. With the exception of the Adirondack anorthosite, initial Sr isotopic ratios (Sr_0) also permit discrimination between enriched ($Sr_0 \geq 0.704$) and depleted ($Sr_0 \leq 0.703$) sources (Table 1). The enriched Sr_0 of the Adirondack anorthosite may be related to its anomalously high $^{18}O/^{16}O$, which has been attributed to isotopic exchange with pervasive metamorphic fluids[16], or to bulk assimilation of metasedimentary units by the anorthosite parental magma[17].

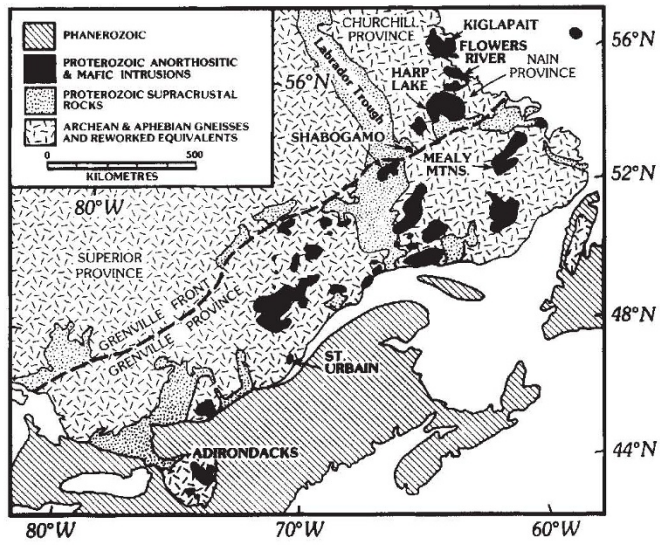


Fig. 1 Geological map of eastern North America showing locations of major Proterozoic anorthosite massifs and mafic intrusions.

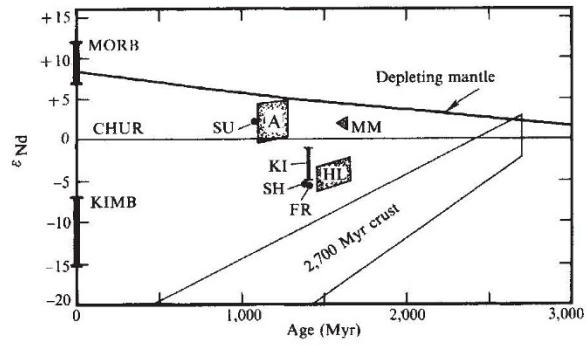


Fig. 2 Plot of age versus ε_{Nd} for Proterozoic anorthositic and mafic intrusions from the eastern Canadian shield. Massifs derived from sources with a depleted signature plot above the chondritic evolution line (CHUR), and include St Urbain (SU), Adirondacks (A) and Mealy Mountains (MM). Intrusions plotting below the CHUR line including Kiglapait (KI), Shabogamo (SH), Flowers River (FR) and Harp Lake (HL) may have been derived from enriched reservoirs. Data for some massifs are shown as polygons because of age uncertainties and contamination effects. Curve representing evolution of depleting mantle is from ref. 15; field showing evolution of 2,700-Myr silicic crust is from unpublished data of J.L.W. Ranges of ε_{Nd} for modern-day MORB[29] and Australian kimberlites (KIMB)[18] are also shown.

It must be considered plausible that the Labrador massifs north-west of the Grenville Front do not reflect the isotopic compositions of their mantle sources, but instead were affected by a 'crustal' contaminant enriched in $^{87}Sr/^{86}Sr$ and depleted in $^{143}Nd/^{144}Nd$. It has been suggested, for example, that at least 20% assimilation of a granitic component by a mantle-derived mafic melt would be required to lower the ε_{Nd} of the Shabogamo gabbro from 0 to −5.3 (ref. 13). Much larger volumes of contaminant would be required if the primary melts were originally derived from a depleted source (Fig. 2). It is not clear that sufficient chemical, petrographic or field evidence exists to corroborate such extensive contamination for the Shabogamo gabbro or the other Labrador massifs. In fact, Morse[11] presents arguments against such effects for the Kiglapait intrusion. Selective contamination models involving fluids, for example, are more plausible, but there is no reason why such effects should be limited to the terrain north of the Grenville Front.

We think it is more likely that the Proterozoic massifs were derived from two distinct reservoirs: one was a depleted source, similar to the modern-day MORB source, the other was either a moderately enriched source, or a roughly chondritic one if

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Table 1 Isotopic compositions of Proterozoic anorthosites and mafic intrusions

	Age (Myr)	Sr_0	ε_{Nd}	Ref.
Depleted source				
Adirondacks, New York	1,288 ± 36*	0.7039–0.7048	+0.4 to +4.1	5, 26
	1,113 ± 10†	0.7040–0.7055	−0.3 to +4.4	
St Urbain, Quebec	1,079 ± 22*	0.7029	+2.2	8, this work
Mealy Mt., Labrador	1,576*		+2.0	7, this work
	1,640 ± 3†	0.7026–0.7032	+1.3 to +2.8	
Enriched (?) source				
Harp Lake, Labrador	1,663 ± 45*	0.7040–0.7066	−5.3 to −2.0	27, 28, this work
	1,460†	0.7040–0.7068	−6.1 to −3.4	
Kiglapait, Labrador	1,416 ± 50*	0.7039–0.7066	−4.9 to −1.1	9–11
Shabogamo, Labrador	1,379 ± 65*	0.7041	−5.3	13, 14
Flowers River, Labrador	1,411 ± 48*		−5.6	12
	1,200 ± 210‡	0.7065		

* From Sm–Nd whole-rock and mineral isochron.
† U–Pb zircon age of cross-cutting granitic intrusion.
‡ Rb–Sr errorchron.

contamination has had a role in altering the isotopic compositions of the Labrador massifs. The possibility of having identified a mid-Proterozoic enriched source is particularly interesting in view of the accumulating evidence for enriched reservoirs in present-day sub-continental settings[18–21]. Both depleted and enriched (?) sources had to be juxtaposed before 1,600–1,650 Myr ago to give rise to the Mealy Mountains and Harp Lake massifs (Table 1). However, both sources must have been formed much earlier, particularly if they developed from a single reservoir with an originally chondritic Sm/Nd. Determination of the ages and extent of these reservoirs beneath the North American continent awaits further isotopic study.

Evidence for a regionally-depleted mantle beneath the Superior Province of North America throughout the Proterozoic comes from Sr and Pb isotopic data for alkaline complexes in Ontario[22,23]. There is also a growing body of Nd isotopic data for the late Archaean rocks of the Superior Province that indicates mantle sources with ε_{Nd} of +1 to +4.5 (ref. 24). A depleted mantle signature is also recorded in middle Proterozoic rocks of the western USA[15]. Therefore, it would almost be expected that the mantle-derived middle Proterozoic rocks of the Grenville Province should have positive ε_{Nd} values. In this context, it is the Nain Province rocks, with their enriched signature, that are unusual. If the isotopic signature of these rocks truly reflects enriched mantle, and not crustal contamination, then this enriched reservoir might consist of: (1) small mantle regions beneath North America that did not participate in Archaean crustal genesis, (2) depleted mantle that was enriched by metasomatic processes shortly after the major Archaean crust formation event or (3) a tapping of a primitive, undepleted source from a deep mantle reservoir[25].

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Post-glacial arid episodes in Ethiopia have implications for climate prediction

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Several lines of geomorphological and stratigraphical evidence suggest that the widespread early post-glacial wet phase in tropical Africa between 12,500 and 5,000 yr BP (refs 1, 2) was interrupted by severe dry episodes during the intervals 11,000–10,000, 8,500–6,500 and 6,200–5,800 yr BP (refs 3–9). The magnitude of the second event has recently been questioned[10]. We present new stratigraphical and dating evidence from the Ziway-Shala lake basin, Ethiopia, indicating the severity and timing of dry phases during the interval 12,000–5,000 yr BP. Our findings have important implications for predictions of the climatic impact of increasing atmospheric levels of CO_2.

The Ziway-Shala Basin (7°–8°30′ N) is a closed drainage in the Main Ethiopian Rift. The four modern lakes are the shrunken remnants of a large late Quaternary palaeolake (2,960 km^2), with an outlet at ~1,670 m above sea level (a.s.l.) into the adjacent Awash catchment (Fig. 1)[11,12]. The lowest lake, Shala, is 266 m deep. Its surface stood at ~1,558 m in 1972, which is referred to as Shala datum (SD).

Fluctuations in water level during the late Quaternary have been reconstructed from several types of evidence: dated shorelines, stratigraphical studies of the rift-floor sediments[12], and analyses of the pollen[13], diatoms[14] and chemistry (F.A.S-P.,

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